

Lattices: Geometry, Hardness, and Reductions

Lecture 5 of 6 — Foundations

SEC 595 — Encrypted Computing

- 1 Gram–Schmidt and Basis Quality
- 2 How Short Is Short?
- 3 The Computational Problems
- 4 q -ary Lattices: The Missing Link

Where we are going

Lectures 1–4 built an algebraic machine: $\mathbb{Z}_q$, the ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, the NTT, modules R_q^k , RNS and gadget decompositions, batching, automorphisms, embeddings.

Every one of those tools is about **computing**.

Not one of them says why anything should be **hard**.

This lecture supplies the missing half, and the hardness is geometric.

Lecture 1 fixed n as the **ring dimension**, the degree of $X^n + 1$. Most lattice texts use n for the lattice rank.

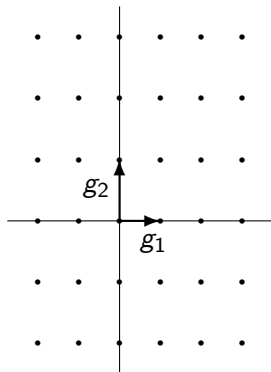
d dimension of the ambient space $\mathbb{R}^d$

k rank of the lattice

n ring dimension (Section 8 only)

Lattices and Their Bases

A lattice is a regular, infinitely repeating grid of points: choose a few arrows, then take every point reachable from the origin by a *whole number* of steps along each arrow, forwards or backwards.



Definition — lattice, basis, rank, dimension

Let $b_1, \dots, b_k \in \mathbb{R}^d$ be linearly independent over $\mathbb{R}$ and $B = (b_1 \mid \dots \mid b_k)$. The **lattice** they generate is

$$\mathcal{L}(B) = \{ Bx : x \in \mathbb{Z}^k \} = \{ x_1 b_1 + \dots + x_k b_k : x_i \in \mathbb{Z} \}.$$

$b_1, \dots, b_k$ is a **basis**, B the **basis matrix**, k the **rank**, d the **dimension**; **full-rank** when $k = d$.

Equivalently: a discrete additive subgroup of $\mathbb{R}^d$.

Example — a basic lattice

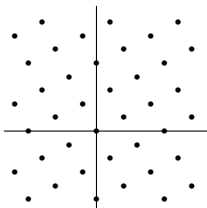
Let $g_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $g_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$, so $G = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $\mathcal{L} = \mathcal{L}(G)$.

- i) Describe $\mathcal{L}$ as a set. Write $(4, -3)$, $(-2, 6)$ and $(6, -9)$ as integer combinations of g_1, g_2 .
- ii) Give a test for whether (u, v) lies in $\mathcal{L}$, and apply it to $(6, -8)$ and $(1, 0)$.
- iii) What do $b_1 = (1, 2)$ and $b_2 = (2, 4)$ generate, and why is that not a rank-2 lattice?

Example — another lattice, not axis-aligned

Let $h_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $h_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, so $H = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $\mathcal{M} = \mathcal{L}(H)$.

- ❶ Write $(3, 4)$, $(4, 2)$, $(0, 5)$ as integer combinations of h_1, h_2 , and mark them on the grid.
- ❷ Compute H^{-1} and use it to decide whether $(1, 1)$ and $(-1, 2)$ lie in $\mathcal{M}$.
- ❸ Find a single congruence $au + bv \equiv 0 \pmod{m}$ that describes $\mathcal{M}$.



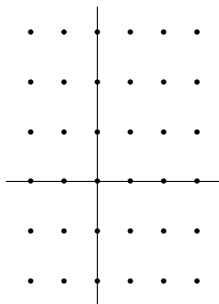
Theorem

Let $B, \tilde{B} \in \mathbb{R}^{d \times k}$ have linearly independent columns. Then $\mathcal{L}(B) = \mathcal{L}(\tilde{B})$ if and only if $\tilde{B} = BU$ for some **unimodular** $U \in \mathbb{Z}^{k \times k}$, i.e. an integer matrix with $\det U = \pm 1$.

Example — one lattice, two bases

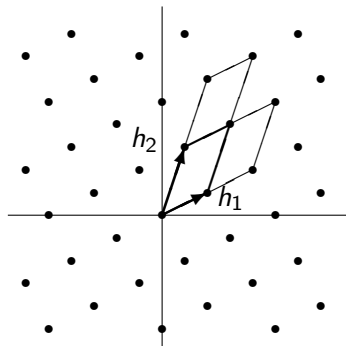
Keep $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$ with basis G , and let $U = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $B = GU$.

Is U unimodular? Compute B ; show its columns lie in $\mathcal{L}$ and that they generate g_1, g_2 back. Draw both bases on the grid.



The Determinant

A lattice tiles space with copies of a single cell, the parallelogram spanned by the basis vectors. The determinant is the area of that cell — the amount of room each lattice point has to itself.



Definition — fundamental parallelepiped and determinant

$$\mathcal{P}(B) = \left\{ \sum_i x_i b_i : x_i \in \mathbb{R}, 0 \leq x_i < 1 \right\},$$

$$\det(\mathcal{L}) = |\det B| = \text{vol}(\mathcal{P}(B)),$$

for full-rank $\mathcal{L} = \mathcal{L}(B)$; for rank $k < d$, $\det(\mathcal{L}) = \sqrt{\det(B^T B)}$.

Proposition

$\det(\mathcal{L})$ does not depend on the basis, since $\tilde{B} = BU$ with $|\det U| = 1$. If $\mathcal{L} \subseteq \mathbb{Z}^d$ is full-rank then $\det(\mathcal{L}) = [\mathbb{Z}^d : \mathcal{L}]$; a lattice cut out of $\mathbb{Z}^d$ by k independent congruences modulo q therefore has $\det(\mathcal{L}) = q^k$.

Section 7 uses that last clause: $\det \Lambda_q^\perp(A) = q^k$.

Example — determinant and index agree

Keep $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$ with the two bases

$$G = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 4 \\ 3 & 9 \end{pmatrix}.$$

Compute $\det(\mathcal{L})$ from each basis. Describe $\mathcal{P}(G)$ and $\mathcal{P}(B)$ geometrically and give their areas.

Then list the cosets of $\mathcal{L}$ in $\mathbb{Z}^2$ directly, and check the two answers against each other.

A basis is *good* when its vectors sit nearly at right angles to one another, and *bad* when they lie nearly along one line.

Gram–Schmidt strips from each vector the part it shares with the earlier ones. How little survives measures how far from orthogonal the basis is.

The **orthogonality defect** compresses that into one number: 1 for a right-angled basis, large for a skewed one.

Definition

For linearly independent $b_1, \dots, b_k$,

$$\tilde{b}_1 = b_1, \quad \tilde{b}_i = b_i - \sum_{j < i} \mu_{i,j} \tilde{b}_j, \quad \mu_{i,j} = \frac{\langle b_i, \tilde{b}_j \rangle}{\langle \tilde{b}_j, \tilde{b}_j \rangle},$$

so $B = \tilde{B}M$ with M upper triangular, 1's on the diagonal and the $\mu_{i,j}$ off it. The **orthogonality defect** of a full-rank basis is

$$\delta(B) = \frac{\prod_{i=1}^d \|b_i\|_2}{\det(\mathcal{L})}.$$

Theorem

For a full-rank lattice with basis $b_1, \dots, b_d$,

$$\det(\mathcal{L}) = \prod_{i=1}^d \|\tilde{b}_i\|_2,$$

and consequently $\delta(B) \geq 1$, with equality exactly when $b_1, \dots, b_d$ are pairwise orthogonal.

Proof.

$B = \tilde{B}M$ with $\det M = 1$, so $|\det B| = |\det \tilde{B}|$; and $\tilde{B}$ has pairwise orthogonal columns, so $|\det \tilde{B}| = \prod_i \|\tilde{b}_i\|_2$. Each $\tilde{b}_i$ is b_i minus a projection, so $\|\tilde{b}_i\|_2 \leq \|b_i\|_2$; multiplying gives $\delta(B) \geq 1$. □

Example — a good and a bad basis, ranked

Both of

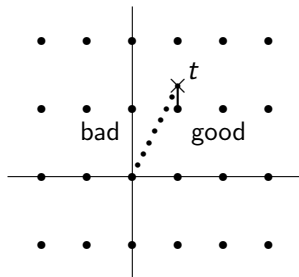
$$g_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, g_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix} \quad \text{and} \quad b_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, b_2 = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

are bases of $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$, with $\det(\mathcal{L}) = 6$.

Compute the orthogonality defect of each.

Then compute the GSO of the second one in exact fractions, and check that the GSO norms multiply to 6.

To decode a point of space to a nearby lattice point: write the point in the basis and round each coefficient to a whole number. A right-angled basis lands on the truly nearest point; a skewed one can land far away.



A **lattice trapdoor** is that asymmetry: public key a bad basis, secret key a good basis of the *same* lattice.

Definition — Babai rounding and the decoding radius

For a basis B of a full-rank $\mathcal{L}$ and a target $t \in \mathbb{R}^d$, **Babai rounding** returns $B \lfloor B^{-1}t \rfloor$; its error lies in $\mathcal{P}(B)$ recentered at the origin.

Babai's **nearest-plane** refinement returns the genuinely closest lattice vector whenever

$$\text{dist}(t, \mathcal{L}) < \frac{1}{2} \min_i \|\tilde{b}_i\|_2,$$

the right-hand side being the **decoding radius** of B .

Example — one target, two bases

Take $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$ and the target $t = (2, 4)$.

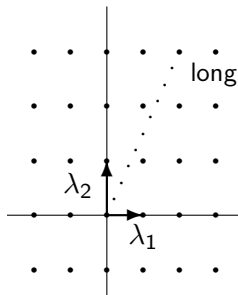
Which lattice point is genuinely nearest t , and how far is it?

Apply Babai rounding with G and then with $B = \begin{pmatrix} 2 & 4 \\ 3 & 9 \end{pmatrix}$. What does each return, and at what distance?

Compute the two decoding radii and explain the difference.

How Short Is Short?

λ_1 is the length of the shortest nonzero vector in the lattice; λ_2 the shortest length at which two independent lattice vectors are available, and so on up.



The solid arrows are shortest of their kind; the dotted one runs to (4, 9).

Definition — successive minima

For $\mathcal{L}$ of rank k and $1 \leq i \leq k$, $\lambda_i(\mathcal{L})$ is the least $r > 0$ such that $\mathcal{L}$ contains i linearly independent vectors of Euclidean norm at most r . So $\lambda_1 \leq \dots \leq \lambda_k$, and vectors attaining the minima need not form a basis.

Theorem — Minkowski

If $S \subseteq \mathbb{R}^d$ is convex, symmetric about the origin, with $\text{vol}(S) > 2^d \det(\mathcal{L})$, then S contains a nonzero lattice point. Applied to cubes,

$$\lambda_1(\mathcal{L}) \leq \sqrt{d} \det(\mathcal{L})^{1/d}.$$

Definition

Assume the lattice points are spread like a Poisson process of density $1/\det(\mathcal{L})$, so a region S holds about $\text{vol}(S)/\det(\mathcal{L})$ of them. Setting that to 1 for a ball of radius r gives

$$\lambda_1(\mathcal{L}) \approx \text{gh}(\mathcal{L}) = \frac{\det(\mathcal{L})^{1/d}}{V_d^{1/d}} \approx \sqrt{\frac{d}{2\pi e}} \det(\mathcal{L})^{1/d},$$

with $V_d = \pi^{d/2}/\Gamma(\frac{d}{2} + 1)$ and $V_d^{1/d} \approx \sqrt{2\pi e/d}$, constant $\sqrt{1/2\pi e} \approx 0.2420$.

It is asymptotic, and unreliable in small dimension.

Example — minima, Minkowski and the heuristic, as numbers

Take $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$ again, with $d = 2$ and $\det(\mathcal{L}) = 6$.

Find λ_1 and λ_2 exactly, and say which vectors attain them.

Evaluate Minkowski's bound, and evaluate the Gaussian heuristic in *both* of its forms.

Which estimates hold, which fail, and by what factor?

Throughout, the input is a basis B of $\mathcal{L}$ — an arbitrary one, which in practice means a bad one.

- **SVP** asks for the shortest nonzero vector of the lattice.
- **CVP** asks, given a point of space, for the lattice vector nearest to it.
- **BDD** is CVP with a promise that the point is already close to the lattice, so the answer is unique.
- **SIS** asks, given a matrix modulo q , for a short nonzero integer vector that the matrix sends to zero.

Definition

Given a basis B of $\mathcal{L}$:

- **SVP**: find nonzero $v \in \mathcal{L}$ with $\|v\|_2 = \lambda_1(\mathcal{L})$.
- **CVP**: given also $t \in \mathbb{R}^d$, find $v \in \mathcal{L}$ minimising $\|t - v\|_2$.
- **BDD**, parameter $\alpha < 1/2$: CVP with the *promise* $\text{dist}(t, \mathcal{L}) < \alpha \lambda_1(\mathcal{L})$, which makes the answer unique.

Definition

γ -SVP nonzero v with $\|v\|_2 \leq \gamma \lambda_1$

γ -CVP v with $\|t - v\|_2 \leq \gamma \text{dist}(t, \mathcal{L})$

SIVP_γ d independent vectors of norm $\leq \gamma \lambda_d$

GapSVP_γ YES if $\lambda_1 \leq r$, NO if $\lambda_1 > \gamma r$ (gap excluded by promise)

Exact SVP and CVP are NP-hard, and SVP stays NP-hard for constant γ ; $\gamma = 2^{O(d)}$ is polynomial-time by LLL (Section 6); at $\gamma = \text{poly}(d)$, where cryptography sits, there is neither an algorithm nor an NP-hardness proof, GapSVP_γ lying in $\text{NP} \cap \text{coNP}$ once $\gamma \geq \sqrt{d/\log d}$.

Definition — SIS, parameters q, k, d, β

Given uniformly random $A \in \mathbb{Z}_q^{k \times d}$ with $d > k$, find nonzero $z \in \mathbb{Z}^d$ with

$$Az \equiv 0 \pmod{q} \quad \text{and} \quad \|z\|_2 \leq \beta.$$

The inhomogeneous version asks for short z with $Az \equiv u$.

Both adjectives are needed: without “short” $z = (q, 0, \dots, 0)$ works, without “nonzero” $z = 0$ works, and β must stay below q . A nonzero z with $\|z\|_\infty \leq 1$ exists by pigeonhole as soon as $2^d > q^k$, i.e. $d > k \log_2 q$; the same counting makes $z \mapsto Az \bmod q$ a compressing map whose collisions are SIS solutions.

Example — a hand-sized SIS instance

Let $q = 13$, $k = 1$, $d = 3$ and

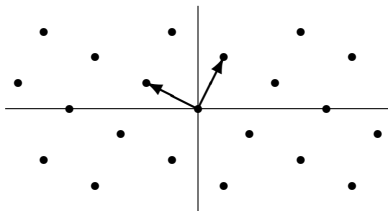
$$A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix} \in \mathbb{Z}_{13}^{1 \times 3}.$$

Find the shortest nonzero $z \in \mathbb{Z}^3$ with $Az \equiv 0 \pmod{13}$. You should be able to spot it without searching.

For which β is this instance solvable, and for which is it trivial?

7.1 Two lattices from one matrix — informally

A matrix A over $\mathbb{Z}_q$ gives two lattices of integer vectors: the ones it *kills*, $Az \equiv 0$, and the ones it *produces*, the integer vectors congruent modulo q to a combination of its rows.



$\Lambda_5^\perp((1 \ 2))$: the pairs with $z_1 + 2z_2 \equiv 0 \pmod{5}$, one point in every five of $\mathbb{Z}^2$, so $\det = 5 = q^k$. The arrows are $(1, 2)$ and $(-2, 1)$, of length $\sqrt{5}$.

Definition — q -ary lattices

For a prime q , integers $k < d$, and $A \in \mathbb{Z}_q^{k \times d}$ of full rank k modulo q ,

$$\Lambda_q(A) = \{ y \in \mathbb{Z}^d : y \equiv A^T s \pmod{q} \text{ for some } s \in \mathbb{Z}^k \},$$

$$\Lambda_q^\perp(A) = \{ z \in \mathbb{Z}^d : Az \equiv 0 \pmod{q} \}.$$

A lattice is **q -ary** when $q\mathbb{Z}^d \subseteq \mathcal{L} \subseteq \mathbb{Z}^d$; both of these are, so both are full-rank in $\mathbb{R}^d$.

Proposition

$\det \Lambda_q(A) = q^{d-k}$ and $\det \Lambda_q^\perp(A) = q^k$; moreover $\Lambda_q^\perp(A) = q \cdot \Lambda_q(A)^*$, which is why attacks come in primal and dual flavours.

7.1 — example: both lattices of one small matrix

Keep $q = 13$, $k = 1$, $d = 3$ and $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix}$.

Write down a basis of $\Lambda_{13}^\perp(A)$ and check its determinant against q^k . What is its λ_1 , and how does the Gaussian heuristic do on it?

Then write down a basis of $\Lambda_{13}(A)$, check its determinant against q^{d-k} , and find its λ_1 .

7.2 An LWE sample is a BDD instance — informally

An LWE sample is a point of the image lattice $\Lambda_q(A)$ nudged by a short error vector.

So recovering the secret means finding the lattice point nearest a target that is *known* to lie close to the lattice.

Proposition

Let $A \in \mathbb{Z}_q^{k \times d}$, $s \in \mathbb{Z}_q^k$, $e \in \mathbb{Z}^d$ short, and $b = A^T s + e \bmod q$. Then b lies within $\|e\|_2$ of $\Lambda_q(A)$, and if

$$\|e\|_2 < \frac{1}{2} \lambda_1(\Lambda_q(A)),$$

the closest lattice point to b is unique and determines s and e .

Search-LWE is BDD on $\Lambda_q(A)$; decision-LWE asks whether b is close to $\Lambda_q(A)$ or uniform in $\mathbb{Z}_q^d$. With $e = 0$ the problem is Gaussian elimination over $\mathbb{Z}_q$.

7.2 — example: a BDD instance you can check by hand

Keep $q = 13$ and $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix}$, so $\Lambda_{13}(A)$ has determinant 169 and $\lambda_1 = \sqrt{17}$.

Take the secret $s = 2$ and the error $e = (1, -1, 1)$. Compute $A^T s$ in centred representatives, then b .

Check the BDD promise. Which lattice point is closest to b , and which is the runner-up? What do s and e come out as?

7.3 SIS is SVP — informally

An SIS solution is a short nonzero vector of the kernel lattice $\Lambda_q^\perp(A)$.

So SIS is the shortest-vector problem with a norm bound in place of exactness.

Proposition

$\{z \in \mathbb{Z}^d : Az \equiv 0 \pmod{q}\}$ is $\Lambda_q^\perp(A)$ by definition, so SIS with parameters (q, k, d, β) is the problem of finding a nonzero vector of norm at most β in $\Lambda_q^\perp(A)$: that is γ -SVP with $\gamma = \beta/\lambda_1(\Lambda_q^\perp(A))$, whose decision version is GapSVP.

	LWE	SIS
Lattice	$\Lambda_q(A)$	$\Lambda_q^\perp(A)$
Determinant	q^{d-k}	q^k
Problem	BDD	SVP with bound β
Short object	the error e	the solution z

7.3 — example: one computation stated twice

With $q = 13$ and $A = (1 \ 5 \ 8)$:

Show that the SIS solution you found earlier and the shortest vector of $\Lambda_{13}^\perp(A)$ are the same vector.

What is γ when $\beta = \sqrt{2}$? When $\beta = 13$?

1. A lattice is a discrete additive subgroup of $\mathbb{R}^d$: the integer span of k independent vectors. Two bases generate it exactly when they differ by a unimodular U .
2. $\det(\mathcal{L}) = |\det B| = \text{vol}(\mathcal{P}(B))$ is basis-independent, equals the index $[\mathbb{Z}^d : \mathcal{L}]$, and is the one invariant every estimate is built from. k congruences modulo q give q^k .
3. Gram–Schmidt makes quality numerical: $\prod_i \|\tilde{b}_i\|_2$ is pinned to $\det(\mathcal{L})$, so the orthogonality defect measures the gap. A good basis kept secret behind a published bad one is a trapdoor.
4. Minkowski bounds $\lambda_1 \leq \sqrt{d} \det(\mathcal{L})^{1/d}$ unconditionally; the Gaussian heuristic is sharper in large dimension and worthless in small.

Summary (2 of 2)

- 5. SVP, CVP and BDD, with factor γ and decision form GapSVP, make a ladder: exact is NP-hard, $\gamma = 2^{O(d)}$ is easy by LLL, and cryptography lives at $\gamma = \text{poly}(d)$ in between.
- 6. SIS asks for short nonzero z with $Az \equiv 0 \pmod{q}$. Both adjectives are load-bearing, and SIS-hardness is collision resistance of $z \mapsto Az$.
- 7. LLL is polynomial; BKZ with block size β interpolates to exact SVP at cost $2^{0.292\beta}$. Quality is the root Hermite factor δ_0 , and choosing (n, q, σ) so the necessary β costs over 2^λ is parameter selection.
- 8. q -ary lattices join the halves: LWE is BDD on $\Lambda_q(A)$, determinant q^{d-k} ; SIS is SVP on $\Lambda_q^\perp(A)$, determinant q^k .
- 9. A rank- k module over a degree- n ring is a lattice of dimension nk , with short vectors in orbits of size n .

Lecture 6 — Noise, Distributions, and the Definitional Toolkit

This lecture said that the error e is the entire source of hardness, and left “short” and “random” as informal words.

Lecture 6 makes them precise: discrete Gaussians, subgaussian tails, the LWE and Ring-LWE definitions, statistical and computational indistinguishability, and the noise-budget arithmetic every FHE correctness proof runs on.